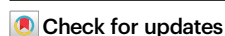


Chemical intuition on bond-dissociation energies as an emergent ability of universal machine-learning interatomic potentials

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Machine-learning interatomic potentials have demonstrated power-law scaling in predictive accuracy as training data and model capacity increase, but it remains unclear whether models trained at scale acquire interpretable chemical concepts. Here we show that an E(3)-equivariant machine-learning interatomic potential learns local bond information without direct supervision of bond properties. To expose this information, we develop an edge-wise emergent energy-decomposition framework and apply it to an Allegro neural-network potential trained on SPICE2, a dataset composed of stable molecular structures. The framework analyzes edge-wise energy contributions obtained from the trained model and their distributions together with information entropy to examine how data size, data composition and model-training scenarios shape the internal representation of chemical bonds. The resulting bond-dissociation energy estimates for archetypal bond types agree quantitatively with literature values and are consistent across models trained on organic and inorganic datasets. We further examine a hybrid training set and find that combining complementary chemical data improves transition-state energy-prediction accuracy while reshaping the learned bond representations. These results indicate that scalable interatomic potentials can acquire transferable bond concepts without explicit bond-energy labels, providing a way to analyze the transition-state energy-prediction problem.

Machine-learning interatomic potentials (MLIPs) and universal machine-learning interatomic potentials (uMLIPs) have advanced computational simulation by delivering near-quantum accuracy at substantially reduced computational cost^{1–3}. Recent progress has been driven primarily by scaling—increasing model size and dataset diversity—leading to predictable power-law improvements in accuracy^{4,5}. This scaling success has enabled software frameworks^{6–13}, also large and diverse datasets including SPICE^{14,15}, MPTrj¹³, Alexandria¹⁶, TM23¹⁷, OMat24¹⁸, OMol25¹⁹, and MatPES²⁰. These advances have produced increasingly generalizable MLIPs for applications ranging from battery materials²¹, catalysts^{22,23}, drug discovery^{24,25}, and nanodevices^{12,26–28}.

While current trends of MLIP development follow a “bigger and more diverse” strategy^{4,5,29}, it has become increasingly clear that the strategy faces challenges in the prediction of complex chemical reactions²². Recent studies have shown that scaling based on conventional loss functions (energy, force, and stress) does not ensure smooth and stable molecular-dynamics (MD) trajectories^{30,31}. To further advance uMLIP development³², a deeper understanding of the nature and characteristics of internal representations (IRs) within uMLIPs is needed^{33,34}. Chorna et al. have compared uMLIPs using the feature reconstruction error³⁵ and found that IRs are stored in a distinct manner. Edamadaka et al. have demonstrated an alignment of IRs

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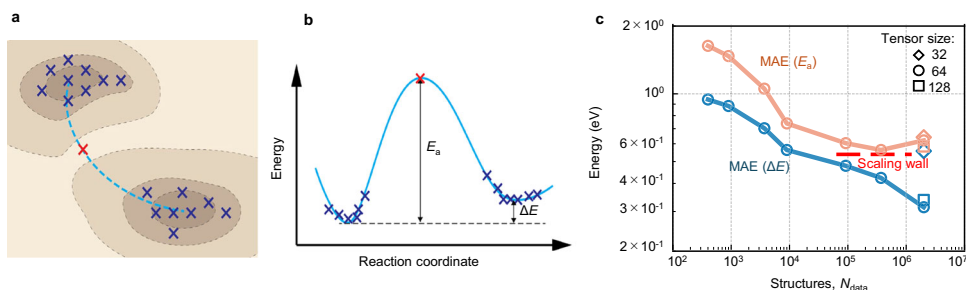


Fig. 1 | Schematic of potential-energy surface learning. **a** Contour plot of a potential energy surface learned by an MLIP. Blue markers indicate sampled data points near the ground states, and red marker indicates near transition state (TS). **b** Energy profile along reaction coordinate. ΔE denotes the reaction energy, and E_a denotes the activation energy. **c** Scaling behavior of prediction errors for ΔE and E_a

across architectures and training-data modalities using 59 MLIP models³⁴.

Accurate prediction of chemical reactions is one of the long-sought capabilities of MLIPs. Figure 1a,b schematically presents the potential energy surface (PES) and reaction pathway learned by an MLIP. A mechanistic understanding of reaction pathways and energy barriers may help accelerate materials development and synthesis. Therefore, many studies have addressed this problem, such as large-scale data-generation initiatives, e.g., the Open Catalyst Project^{22,23,36}, and active learning for reaction modeling including ANI potential²⁹. Recent development in foundation models has also demonstrated predictive capability for chemical reactions. These models have shown improved performance in chemical reaction prediction through training on a hybrid dataset that combines organic and inorganic materials data^{37,38}.

Despite these advancements, it remains challenging to interpret how models learn IRs relevant to chemical bonding and reactions during training. For example, Fig. 1c illustrates a disparity in the scaling behavior of reaction energy (ΔE) and activation energy (E_a) using Allegro³⁹ models trained on SPICE2 dataset¹⁴ and evaluated on Transition1x (T1x) dataset⁴⁰. The accuracy of ΔE improves consistently with increasing training data in all tensor sizes, consistent with previously reported scaling behavior. In contrast, the predicted E_a plateaus around the data size of 10^5 , hitting a “scaling wall” where additional data provides little to no benefit. This disparity is observed consistently across larger model sizes, indicating limitations beyond model capacity alone.

This observation of the scaling disparity has given rise to several scientific questions: Why does the model learn to predict ΔE while struggling with E_a ? The training is performed solely on SPICE2 dataset¹⁴, which consists primarily of stable molecular structures and includes isolated atoms, small radicals, and molecules spanning neutral, cationic, and anionic charge states; however, it does not contain any transition-state (TS) geometries. At the same time, we observe that the prediction accuracy of E_a continues to improve as the size of training set grows to $\approx 10^5$. While the scaling behavior of E_a is somewhat expected, elucidating the underlying mechanism of the distinct scaling behaviors may help further advance MLIP development. Are there underlying data representations that could explain the learning behaviors? Is there a metric to capture the mechanisms behind these scaling disparities and help quantify chemical information that is internally learned by an MLIP?

To address these questions, we propose an edge-wise emergent energy-decomposition (E3D) framework based on the chemical bond and its associated bond-dissociation energy (BDE)⁴¹. Unlike the conventional approach, where the total energy is defined as the sum of atomic energies, we consider the total energy as the sum of bond energies. Because the bond energy does not explicitly appear in the

on the Transition1x (T1x) dataset⁴⁰. Mean absolute error (MAE) is plotted as a function of the number of SPICE2¹⁴ training structures. Diamonds, circles and squares indicate Allegro tensor sizes 32, 64, and 128, respectively. While ΔE accuracy improves consistently, E_a predictions plateau, indicating a scaling wall.

loss function, it provides a way to assess whether a model has learned relevant chemical information.

E3D further leverages a set of IRs, i.e., the symmetric (D_{ij}) and asymmetric (A_{ij}) terms given in Eq. (7). The two-dimensional probability distribution of D_{ij} vs. A_{ij} and associated Shannon entropy H_2D enable quantitative assessment of training progress and provide a way to track when and how an MLIP learns bond-related representations during training.

In this study, we employ the E(3)-equivariant Allegro architecture³⁹ and use E3D to characterize bond-related information learned by the model. We also apply E3D to “the unreasonable effectiveness of hybrid data”⁴², where TS energy prediction may be improved by hybridizing organic and inorganic materials datasets^{37,38}. We evaluate hybrid training and discuss its relationship to the scaling wall problem. Additional methodological details, dataset summaries, and validation analyses are provided in the Supplementary Information (Supplementary Figs. 1–3 and Supplementary Tables 1–6).

The main contributions of this study are as follows. First, we show that a scalable model trained on large materials datasets learns consistent bond-type-dependent representations, and that this behavior is observed across different materials datasets. Second, we show that probability distributions of IRs and their information entropy provide a tool for qualitative and quantitative evaluation across model-training scenarios. Third, we examine how a hybrid dataset affects TS energy prediction and discuss its implications for the scaling wall during training.

Results

Emergent bond-dissociation energy

First, we demonstrate that the Allegro model³⁹ can acquire knowledge of BDE without explicit supervision. Specific constraints were imposed on Allegro to achieve output decomposition as described in Eq. (4). We trained the Allegro model using SPICE2¹⁴ and evaluated the generalization performance using the T1x⁴⁰ dataset. Subsequently, we analyzed changes in IRs with increasing amounts of training data and verified that this monitoring functioned appropriately.

Figure 2a compares D_{ij} against BDEs from literature. The model generates a distribution that aligns well with established chemical trends. Not only is the order of bond energies for C-H, N-H, and O-H correctly predicted, but the average bond energies also show quantitative agreement with the reference BDE. The model also appears to have learned the concept of bond order correctly, for example, single, double, and triple bonds for C-C, C-N, C-O, and N-N bonds, despite the fact that the model was trained solely on the total system energy. We refer to this model capability as ‘emergent BDE’.

BDE is a thermodynamic quantity that is often evaluated by comparing the enthalpy difference before and after bond breaking⁴³,

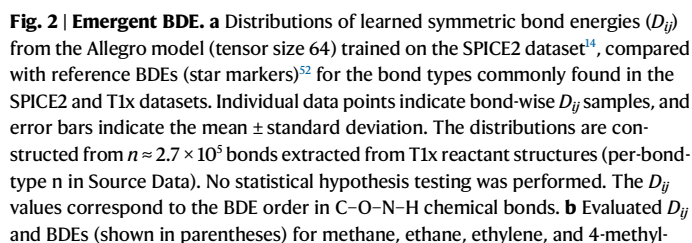


Figure 1 is a scatter plot titled "Literature BDE" showing the Bond Dissociation Energy (BDE) in eV for various bond types. The y-axis represents Energy (eV) from -10 to 0. The x-axis represents Bond type, categorized by element pairs: C-H, N-H, O-H, C-C, C=C, C≡C, C-N, C=N, C=O, C≡O, N-N, N=O, and N≡O. Data points are color-coded by element: C (orange), N (green), O (blue), and C-N (purple). Red stars represent literature values. The plot shows a general trend where BDE decreases as the bond order increases for a given pair of elements, with C-H having the highest BDE and N≡O having the lowest.

Figure 4a shows the distribution of the IRs, the symmetric D_{ij} and asymmetric A_{ij} terms given by Eq. (7). We have examined the distributions over archetypal chemical bond types (C-H, C-C, C-O, C-N) across varying training conditions. Here, each training set with different size is randomly sampled from the entire SPICE2 dataset. The distribution function shows progressive development and structural refinement that evolve into isolated peaks as the training size increases. This result indicates increasing development of complex and

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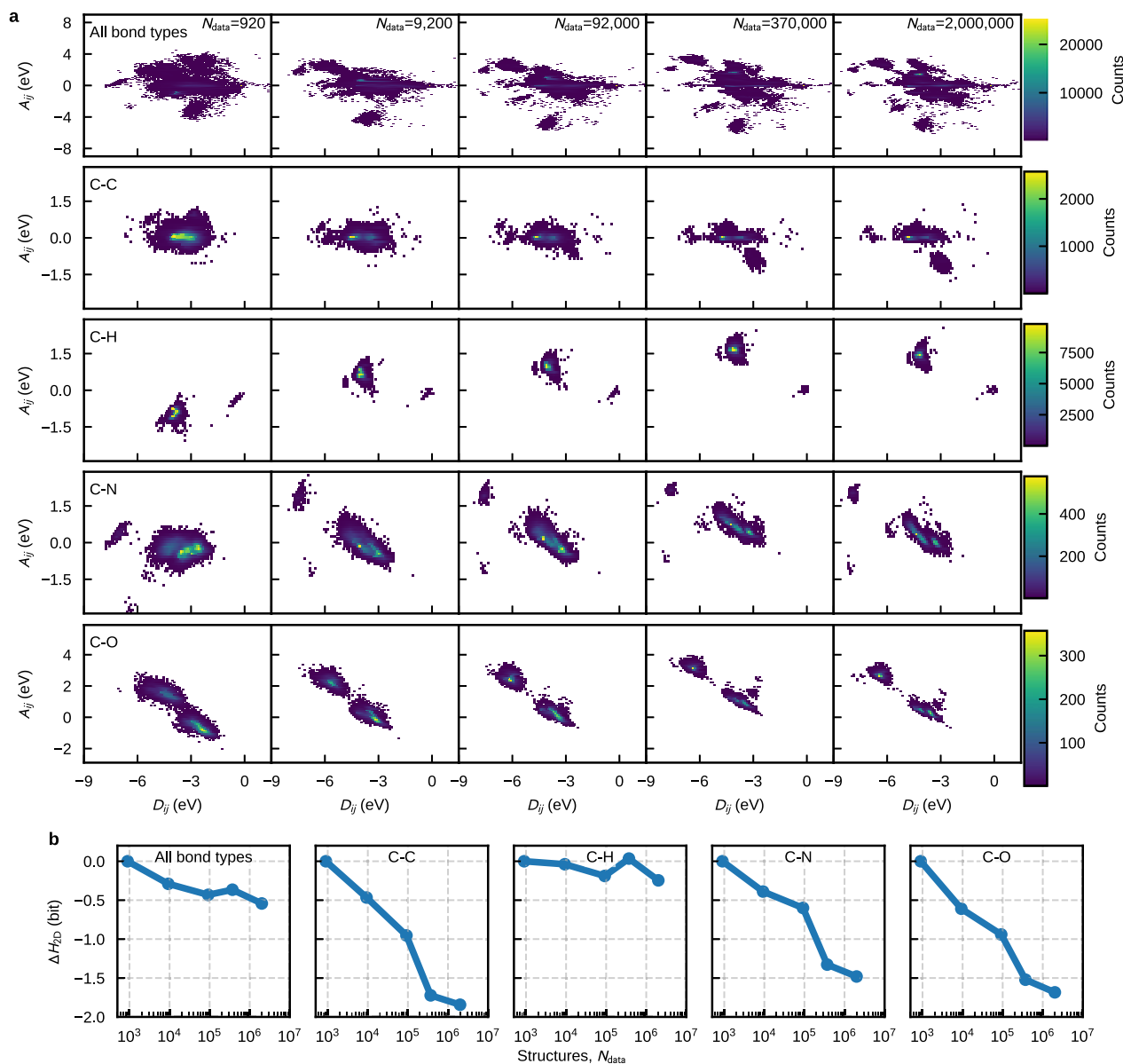


Fig. 4 | Evolution of bond-energy representations. **a** IR distributions for C-C, C-H, C-O, and C-N bonds at varying training-data volumes N_{data} . Here, we employ the structure of the test set, which has been separated from the training set in SPICE2.

b H_{2D} plot as a function of N_{data} , in which the entropy value at $N_{\text{data}} = 920$ as reference. As N_{data} increases, more refined structures and boundaries develop, resulting in the consistent reduction of H_{2D} .

multi-modal features inside the model. Each island of the distribution corresponds to distinctive bond characteristics such as bond order; therefore, the obtained IR distribution may be seen as a fingerprint of a given atomic pair.

Using H_{2D} , we also quantify the changes in the distribution during training as a function of the training-set size. Figure 4b shows consistent decreasing trends in H_{2D} with respect to the size of training set. The entropy computed from the combined distribution of all examined bond types converged at $\approx 10^5$ – 10^6 training structures, as shown in Fig. 4b. Bond-resolved analysis reveals distinct learning behaviors. For C-C, C-N, and C-O bonds, ΔH_{2D} steadily decreases between $N_{\text{data}} = 10^3$ and 10^5 , with an approximate slope of -0.5 with respect to $\log(N_{\text{data}})$. The trend then substantially slows down for $N_{\text{data}} > 10^5$. In contrast, ΔH_{2D} of the C-H bond remains essentially flat, indicating convergence even with $N_{\text{data}} < 10^3$. These results suggest greater complexity in the C-C, C-N, and C-O bond distributions than in the C-H bond distribution.

Hybrid dataset and TS energy prediction

Lastly, we examine the effect of the hybrid dataset on TS energy prediction. We trained the models on either SPICE2 alone or a hybrid dataset consisting of SPICE2 + MatPES (labeled as Hybrid).

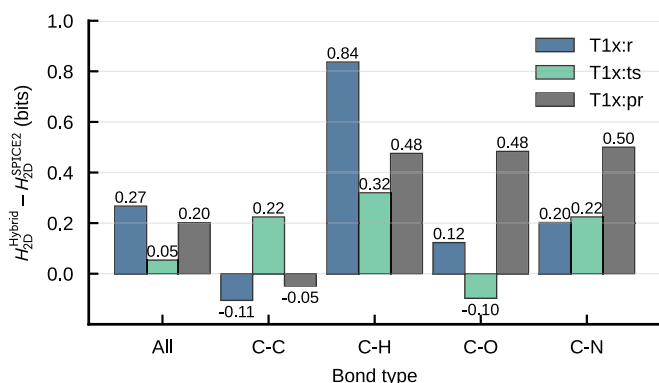
Table 1 summarizes the model performance using mean absolute error (MAE) of E_a and ΔE . As previously reported, we have also confirmed that the performance of TS energy prediction improves consistently using the hybrid dataset regardless of the tensor size, for example, achieving an E_a MAE of 0.44 eV (Hybrid) compared to 0.58 eV (SPICE2).

Entropy on Transition1x subsets

Figure 5 shows the breakdown of H_{2D} by bond types tested on the T1x dataset. Here we divide T1x into three groups, that is, the reactant structures (labeled T1x:r), TS structures (labeled T1x:ts), and the product structures (labeled T1x:pr). We found that entropy tended to increase across various bond types, particularly in C-H bonds across all T1x groups.

Table 1 | Performance improvements with hybrid dataset training on T1x reactive properties

Dataset	Tensor size	E_a MAE (eV)	ΔE MAE (eV)
SPICE2	64	0.61	0.31
	128	0.58	0.33
Hybrid	64	0.49	0.27
	128	0.44	0.25

**Fig. 5 | Entropy change in IR distributions using models trained on SPICE2 and Hybrid (SPICE2 + MatPES).** The entropy is evaluated using the T1x dataset, which is divided into three mutually exclusive sets of atomic configurations: reactant structures (T1x:r), TS structures (T1x:ts), and product structures (T1x:pr). Four types of chemical bond (C–H, C–C, C–O, and C–N) are examined here. Entropy increases in various chemical bonds. C–H bonds show this across all T1x datasets. C–O and C–N bonds show especially high entropy increases in T1x:pr.

Discussion

The E3D framework provides a way to analyze how BDE representations develop within MLIPs. The emergent BDE may help explain MLIP generalizability because an MLIP that captures basic chemical building blocks can, in principle, support stable simulations of large molecular systems (e.g., proteins)⁵. Currently, most MLIPs are trained on system energy, atomic force, and system stress; therefore, other scalable MLIPs^{10–12} may have acquired similar IRs. Representative molecular-dynamics stability trajectories for crystalline silicon and aspirin are provided as Supplementary Movies 1 and 2.

The emergent BDE is achieved neither with a specific loss function nor inductive bias but is learned from data. Unsupervised BDE learning, therefore, leaves room for improved prediction accuracy. For instance, supervised BDE models such as BondNet⁴¹ have achieved a prediction error of 0.022 eV, while the emergent BDE differs from literature values by about 0.3 eV. Here, two approaches may be considered: (1) explicit BDE supervision in the loss function, and (2) further increasing the size of training data. For example, OMat24¹⁸ and OMol25¹⁹ contain over 100 million structures, which is over two orders of magnitude greater than the dataset used in this study. While evaluating whether these approaches outperform existing BDE models is beyond the scope of this study, it is useful to assess them as model and dataset scales increase⁴⁵.

We found that the emergent BDE is observed with the two widely-used large-scale materials datasets, SPICE2 and MatPES. This result suggests that the observed behavior is not specific to the SPICE2 dataset that predominantly consists of the bond types we examined, and is also present in the MatPES-trained model.

The IR distribution and information entropy provide insights into MLIP development and enable us to inspect the process from multiple aspects. By changing the data size, the distributions exhibit the formation of domains that go through the refinement of their boundaries and the shrinkage in size. In addition, such domains can split into

further smaller clusters as the training set increases (see Fig. 4a). The result indicates that a better understanding of the local atomic environment and bond energies has developed inside the model.

The change in entropy with respect to the size of the training set provides insight into model training and correlates with model scalability. It is useful for comparing several training scenarios, for example, SPICE2 versus MatPES or different training-set sizes.

A declining trend shown in Fig. 4b suggests that the IR distribution converges to narrower peaks with a smaller variance. How many peaks (possible BDE values) exist depends on the training set, for example, the C–C bond with different multiplicity. The value of entropy could remain the same when the distribution barely changes (for example, C–H bond in Fig. 4a) or increases if more complex structures emerge as model training proceeds.

We found that the hybrid dataset exceeds the scaling wall with the SPICE2 dataset on the prediction of TS energy, achieving a further reduction of the MAE loss by $\approx 25\%$ for E_a and ΔE . The increased entropy suggests the formation of more complex multi-modal IR structures than the one with SPICE2 alone. Material datasets that consist of molecules or bulk systems usually have very different characteristics due to their symmetries, boundary conditions, and underlying quantum mechanics theory used to generate training set; therefore, a common practice is to train separate MLIPs depending on target applications.

E3D analysis indicates that SPICE2–MatPES hybrid training improves out-of-distribution performance for TS energy prediction. Although the mechanisms underlying these gains remain to be understood, several studies have reported similar observations^{46–49}.

In summary, we have developed the E3D framework that provides a mechanism-based perspective on the generalization of universal MLIPs. Current MLIP advancements largely rely on methodologies designed for generic AI models; a physically inspired method such as E3D may help guide analyses of model scaling and generalization limits.

Methods

N-body interaction

The potential energy E_{system} of a system composed of N atoms can be described using a body-order expansion that includes terms up to the N -body interaction:

$$E_{\text{system}} = \sum_i E_i^{(1)} + \sum_{ij} E_{ij}^{(2)} + \sum_{ijk} E_{ijk}^{(3)} + \dots + (N - \text{body term}). \quad (1)$$

Since the computational cost of N -body interactions scales as $O(N^N)$, directly incorporating such terms into MLIPs is not practical. Many of the current MLIPs allow the E_{system} to be expressed as a sum of single-atom energies $\tilde{E}_i^{(1)}$ that include many-body effects:

$$E_{\text{system}} = \sum_i \tilde{E}_i^{(1)}. \quad (2)$$

To accomplish this, the existing MLIPs, such as Behler–Parrinello neural-network potentials¹, describe many-body interactions by combining two-body and three-body descriptors through nonlinear functions, such as activation functions used in neural networks. In particular, atomic cluster expansion (ACE)⁹, which enables the construction of many-body descriptors in a computationally efficient manner, has improved the accuracy of the representation in Eq. (2).

However, it is not necessary to represent the total energy with the single-atom energy $\tilde{E}_i^{(1)}$. Here, the cohesive energy is defined as

$E_{\text{coh}} \equiv E_{\text{system}} - \sum_i E_i^{(1)}$, and then Eq. (1) becomes,

$$E_{\text{coh}} = \sum_{ij} E_{ij}^{(2)} + \sum_{ijk} E_{ijk}^{(3)} + \dots + (N - \text{body term}). \quad (3)$$

E_{coh} is preferred to E_{system} for the training target of MLIPs, because it eliminates dependencies on computational settings such as pseudopotentials in first-principles calculations, thereby providing an unbiased energy reference.

Similar to that the representation from Eq. (1) to Eq. (2) was achieved, MLIPs should be able to expand E_{coh} in Eq. (3) using two-body interactions $\tilde{E}_{ij}^{(2)}$ that include many-body effects, as follows:

$$E_{\text{coh}} = \sum_{ij} \tilde{E}_{ij}^{(2)}. \quad (4)$$

Since $\tilde{E}_{ij}^{(2)}$ can be naturally interpreted as a quantitative measure of the attractive or repulsive interatomic interactions, Eq. (4) provides a basis for evaluating interatomic bond strengths using MLIPs themselves.

Model and energy-decomposition formulation

We employ the E(3)-equivariant Allegro architecture³⁹ for our analysis due to its inherent energy decomposability shown in Eq. (4). We trained Allegro models with internal tensor representation sizes of 32, 64, and 128, spherical harmonic expansion $l_{\text{max}} = 2$, and radial cutoff $r_{\text{cut}} = 5.2$ Å. For most analysis, we employed a tensor size of 64, prioritizing computational efficiency without compromising accuracy. This specific size proved optimal: a smaller tensor (size 32) resulted in lower accuracy due to its limited parameter count and thus reduced representational capacity, whereas a larger tensor (size 128) demonstrated comparable performance to size 64.

Allegro decomposes total system energy E_{system} into per-node (atom) energies ε_i and per-edge energies ε_{ij} ³⁹:

$$\begin{aligned} E_{\text{system}} &= \sum_{i=1}^N (\sigma_{Z_i} \varepsilon_i + \mu_{Z_i}), \\ \varepsilon_i &= \sum_{j \in N(i)} \sigma_{Z_i Z_j} \varepsilon_{ij}, \end{aligned} \quad (5)$$

where σ_{Z_i} , $\sigma_{Z_i Z_j}$ are learnable per-species scale parameters, μ_{Z_i} are learnable shift parameters for atoms of species Z_i and Z_j , N is the total number of atoms, and $N(i)$ represents atom i 's local neighborhood. The Allegro model, which is based on the ACE formalism⁹ and combined with their connection via nonlinear functions, enables the expression of E_{system} in terms of the per-edge energies ε_{ij} ³⁹.

Therefore, to implement an architecture that satisfies Eq. (4) within the Allegro model, we standardize the normalization parameters by setting the shift parameters $\mu_{Z_i} = 0$ and scale factors $\sigma_{Z_i} = \sigma_{Z_i Z_j} = 1.0$ in Eq. (5). Furthermore, instead of learning E_{system} , the model is trained to predict E_{coh} . With these settings, Eq. (5) becomes a direct sum of edge energies ε_{ij} :

$$E_{\text{coh}} = \sum_{i=1}^N \sum_{j \in N(i)} \varepsilon_{ij}, \quad (6)$$

which is equivalent to Eq. (4).

Importantly, ε_{ij} and ε_{ji} are generally not the same (i.e., $\varepsilon_{ij} \neq \varepsilon_{ji}$) reflecting the local environments of i and j atoms independently. Thus, these energy values differ based on which atom index to be taken as the central atom. We define a set of metrics, i.e., the symmetric component D_{ij} and asymmetric component A_{ij} , to capture the degree of the

bond-energy strength and its asymmetry:

$$\begin{aligned} D_{ij} &= \varepsilon_{ij} + \varepsilon_{ji}, \\ A_{ij} &= \varepsilon_{ij} - \varepsilon_{ji}. \end{aligned} \quad (7)$$

Here, D_{ij} represents total bond energy between i th and j th atoms. The summation of D_{ij} for all atomic pairs in the system, i.e., $\sum_i \sum_{j \in N(i), j > i} D_{ij}$ is equal to E_{coh} . The asymmetric component A_{ij} quantifies energy imbalance arising from the difference in the local environment of the two atoms. For example if i th and j th atoms are of the same element, A_{ij} reflects the difference between the two local environments.

Model training

All models were trained using the NequIP framework¹⁰ with the Allegro architecture³⁹. For models with tensor size 64, the two-body latent multilayer perceptron (MLP) had dimensions [64, 128, 256], the later latent MLP had dimensions [256, 256, 256], and the final edge-energy MLP used one hidden layer with dimension 128. All models used three interaction layers. The models were optimized using Adam with a learning rate of 1×10^{-3} . Batch sizes were varied between 8 and 128 depending on the training-set size. Training was continued until convergence according to the validation loss, with at least 100 epochs for each training run.

The model parameters were optimized by minimizing a weighted sum of root-mean-square errors for per-atom cohesive energy, atomic forces and, for periodic systems, stress:

$$\mathcal{L} = \lambda_E \left\| \frac{E_{\text{coh}} - E_{\text{pred}}}{N} \right\|^2 + \lambda_F \| \mathbf{F}_{\text{ref}} - \mathbf{F}_{\text{pred}} \|^2 + \lambda_S \| \boldsymbol{\sigma}_{\text{ref}} - \boldsymbol{\sigma}_{\text{pred}} \|^2, \quad (8)$$

where N is the number of atoms, $E_{\text{pred}} = \sum_i \sum_{j \neq i} \varepsilon_{ij}$ is the predicted cohesive energy, $\mathbf{F}_{\text{ref}}$ and $\boldsymbol{\sigma}_{\text{ref}}$ are the reference forces and stress tensor, and $\mathbf{F}_{\text{pred}}$ and $\boldsymbol{\sigma}_{\text{pred}}$ are the corresponding predicted quantities obtained by automatic differentiation. The stress term was used only for periodic systems. The loss weights were set to $\lambda_E: \lambda_F: \lambda_S = 1: 1: 1$ for SPICE2 and to $\lambda_E: \lambda_F: \lambda_S = 1: 1: 1$ for MatPES and the hybrid SPICE2 + MatPES dataset.

Datasets and pre-processing

The molecular training data were taken from SPICE2¹⁴, which contains stable molecular configurations. Inorganic training data were taken from the r^2 SCAN subset of MatPES²⁰. The hybrid dataset was constructed by combining SPICE2 and MatPES. For molecular structures from SPICE2 used in hybrid training, the molecules were placed in a cubic cell with a side length of 500 Å and assigned zero target stress.

T1x⁴⁰ was used as an out-of-distribution benchmark for reactive properties. For the T1x reactant, TS and product structures, energies and forces were recalculated using Psi4⁵⁰ at the same level of theory as SPICE2, ω B97M-D3(BJ)/def2-TZVPPD.

E3D protocol

We assigned bond multiplicities (single, double, triple, aromatic) using general Amber force field (GAFF) atom types from antechamber in AmberTools 23⁵¹, followed by manual validation for ambiguous cases. For aromatic bonds, 0.5 was added to the assigned bond multiplicity for the E3D analysis. Reference BDE values from experimental measurements⁵² and established theoretical values⁵³ enable quantitative comparison with learned D_{ij} distributions, providing a direct comparison between the learned representation and reference bond energetics.

In addition we define two key metrics, the BDE distribution shift ΔE_{BDE} and the BDE distribution width σ_{BDE} , to enable a unified evaluation metric across different model training settings. ΔE_{BDE} and σ_{BDE} are computed from a single distribution combining the BDE

Table 2 | Key metrics in E3D framework

Metric	Symbol	Description
Symmetric bond energy	D_{ij}	Total interaction energy: $\varepsilon_{ij} + \varepsilon_{ji}$
Asymmetric bond energy	A_{ij}	Energy imbalance due to asymmetry: $\varepsilon_{ij} - \varepsilon_{ji}$
BDE distribution shift	ΔE_{BDE}	Mean deviation from reference BDEs
BDE distribution width	σ_{BDE}	Standard deviation of D_{ij} distributions
2D map information	$H_{2\text{D}}$	Shannon entropy of the D_{ij} vs. A_{ij} map entropy

distributions of all bond types, each of which is shifted by their reference BDE value (see inset in Fig. 2).

To analyze correlations between the symmetric (D_{ij}) and asymmetric (A_{ij}) components of learned bond energies, we generated 2D histograms of D_{ij} versus A_{ij} . This analysis focused on C–C, C–H, C–O, and C–N from T1x dataset structures, considering covalent bonds shorter than 2.2 Å to ensure chemical relevance. We constructed these histograms using uniform 0.1 eV energy bin widths for both axes over their observed ranges.

Shannon entropy quantifies organization structure of representational spaces:

$$H_{2\text{D}} = - \sum_{k,l} p_{kl} \log p_{kl} \quad (9)$$

where p_{kl} is the normalized frequency in bin (k, l) of the 2D histogram. Lower entropy values indicate more organized, well-defined IRs, suggesting that the model has developed structured chemical understanding. This metric provides a quantitative measure of how clearly the model distinguishes different chemical environments and structure.

Table 2 summarizes the metrics employed in the E3D framework to characterize learned representations and their relationship to physical quantities. Numerical source data for the quantitative main-text figures and tables are provided in the Source Data file.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available from the corresponding author upon request. The publicly available datasets used in this study are available from their original sources: SPICE2¹⁴, MatPES²⁰, and Transition1x⁴⁰. The hybrid dataset used in this work was constructed by combining SPICE2 and MatPES. Unprocessed data, including raw bond-wise edge-energy data used for the E3D analysis, are provided in Supplementary Data 1. Representative MD trajectories used for the stability tests are provided as Supplementary Movies 1, 2. The third-party datasets listed above are subject to the access terms and licenses of their original providers; we imposed no additional restrictions. Source data are provided with this paper.

Code availability

The code used in this work is available from Zenodo⁵⁴. The implementation used PyTorch version 2.5.1, NequIP¹⁰ version 0.6.1, and Allegro³⁹ version 0.3.0.

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Author contributions

S.H., K.S., and K.N. designed the project and wrote the manuscript. K.S. developed the basic method for MLIP energy decomposition. S.H. developed the modified Allegro model and code, performed MLIP training with varying data volumes, and conducted data analyses. K.N. supervised the project. A.N., R.K.K., and P.V. contributed to the discussion and interpretation of the results. All authors contributed to the writing and editing of the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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